

CLASS AND METHODS

Q1. What is an object in Java?

- a) Blueprint of class
- b) Instance of a class
- c) Method definition
- d) Package

Answer: b

Q2. Which keyword is used to create an object?

- a) class
- b) new
- c) create
- d) object

Answer: b

Q3. Which of the following is true about classes?

- a) Class is an instance of object
- b) Class is a blueprint of objects
- c) Class stores only methods
- d) Class cannot have variables

Answer: b

Q4. What is the default value of an instance variable of type int?

- a) 0
- b) null
- c) undefined
- d) garbage value

Answer: a

Q5. Which method is the entry point of a Java program?

- a) start()
- b) run()
- c) main()
- d) init()

Answer: c

Q6. What is the return type of main method?

- a) int
- b) void
- c) String
- d) None

Answer: b

Q7. What does String[] args represent?

- a) Array of integers
- b) Command-line arguments
- c) Object references
- d) Class names

Answer: b

Q8. Which of the following is NOT a method signature component?

- a) Method name
- b) Parameter list
- c) Return type
- d) Parameter types

Answer: c

Q9. What is method overloading?

- a) Same method with same parameters
- b) Same method name with different parameters
- c) Different methods with same return type
- d) Same method in different classes only

Answer: b

Q10. Which of the following is compile-time polymorphism?

- a) Method overriding
- b) Method overloading
- c) Inheritance
- d) Encapsulation

Answer: b

Q11. What is a constructor?

- a) Method returning object
- b) Special method used to initialize object
- c) Static method
- d) Destructor

Answer: b

Q12. What is the name of constructor same as?

- a) Method
- b) Class
- c) Package
- d) Object

Answer: b

Q13. What happens if no constructor is defined?

- a) Compilation error
- b) JVM creates default constructor
- c) Runtime error
- d) Program crashes

Answer: b

Q14. Which constructor has no parameters?

- a) Parameterized constructor
- b) Default constructor
- c) Copy constructor
- d) Static constructor

Answer: b

Q15. Can constructors be overloaded?

- a) Yes
- b) No
- c) Only once

d) Only static

Answer: a

Q16. What is the use of 'this' keyword?

- a) Refers to current class object
- b) Refers to parent class
- c) Refers to static variables
- d) Refers to method

Answer: a

Q17. Which access modifier gives maximum accessibility?

- a) private
- b) protected
- c) public
- d) default

Answer: c

Q18. Which access modifier is accessible only within the class?

- a) public
- b) private
- c) protected
- d) default

Answer: b

Q19. What is default access modifier?

- a) private
- b) public
- c) package-private
- d) protected

Answer: c

Q20. Which memory is used for object storage?

- a) Stack
- b) Heap
- c) Register
- d) Cache

Answer: b

Q21. What is garbage collection?

- a) Memory allocation
- b) Memory cleanup of unused objects
- c) Method execution
- d) Code compilation

Answer: b

Q22. Which method is called before object is destroyed?

- a) delete()
- b) destroy()
- c) finalize()
- d) remove()

Answer: c

Q23. What triggers garbage collection?

- a) Programmer
- b) JVM automatically
- c) OS only
- d) Compiler

Answer: b

Q24. What will be output?

```
class Test {  
    int x;  
    public static void main(String[] args) {  
        Test t = new Test();  
        System.out.println(t.x);  
    }  
}
```

- a) Garbage value
- b) 0
- c) null
- d) Error

Answer: b

Q25. What will be output?

```
class Test {  
    void show() {  
        System.out.println("Hello");  
    }  
    public static void main(String[] args) {  
        Test t = new Test();  
        t.show();  
    }  
}
```

- a) Hello
- b) Error
- c) Nothing
- d) Null

Answer: a

Q26. What will happen if constructor is private?

- a) Object cannot be created outside class
- b) Program crashes
- c) Constructor not called
- d) JVM ignores it

Answer: a

Q27. What will be output?

```
class Test {  
    Test() {  
        System.out.println("Constructor");  
    }  
    public static void main(String[] args) {  
        Test t = new Test();  
    }  
}
```

- a) Constructor
- b) Error
- c) Null
- d) Nothing

Answer: a

Q28. What is recursion?

- a) Loop execution
- b) Method calling itself
- c) Multiple classes
- d) Constructor call

Answer: b

Q29. What is base condition in recursion?

- a) Infinite loop
- b) Condition to stop recursion
- c) Start of method
- d) End of program

Answer: b

Q30. What will be output?

```
class Test {  
    static int fun(int n) {  
        if(n == 0) return 0;  
        return n + fun(n - 1);  
    }  
    public static void main(String[] args) {  
        System.out.println(fun(3));  
    }  
}
```

- a) 3
- b) 6
- c) 9
- d) Error

Answer: b